

1.1. Rozepíšeme z definice a za pomoci axiomu SSS, SL a z pozorování 8.15. (2), (3)

$$\begin{aligned}\|u+v\|^2 - \|u-v\|^2 &= \langle u+v, u+v \rangle - \langle u-v, u-v \rangle = \langle u+v, u \rangle + \langle u+v, v \rangle - (\langle u-v, u \rangle - \langle u-v, v \rangle) = \\ &= \langle u, u \rangle + \langle v, u \rangle + \langle u, v \rangle + \langle v, v \rangle - (\langle u, u \rangle - \langle v, u \rangle - \langle u, v \rangle + \langle v, v \rangle) = \\ &= 2 \langle v, u \rangle + 2 \langle u, v \rangle = \\ &= 2 \overline{\langle u, v \rangle} + 2 \langle u, v \rangle = 2 (\overbrace{\langle u, v \rangle + \langle u, v \rangle}^{2\operatorname{Re}}) = 4 \operatorname{Re}(\langle u, v \rangle)\end{aligned}$$

vlastnost kompl. č. $\left\{ \begin{array}{l} \langle u, v \rangle + \overline{\langle u, v \rangle} = 2\operatorname{Re} \\ x + \bar{x} = 2\operatorname{Re} \\ \swarrow \\ a+bi + a-bi = 2a \end{array} \right.$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

Zde analogicky:

$$\begin{aligned}\|u-iv\|^2 - \|u+iv\|^2 &= \langle u-iv, u-iv \rangle - \langle u+iv, u+iv \rangle = \langle u-iv, u \rangle - i \langle u-iv, v \rangle - (\langle u+iv, u \rangle + i \langle u+iv, v \rangle) = \\ &= \langle u, u \rangle + i \langle v, u \rangle - i \langle u, v \rangle + \langle v, v \rangle - (\langle u, u \rangle - i \langle v, u \rangle + i \langle u, v \rangle + \langle v, v \rangle) = 2i \langle v, u \rangle - 2i \langle u, v \rangle = \\ &= 2i \overline{\langle u, v \rangle} - 2i \langle u, v \rangle = -2i (\langle u, v \rangle - \overline{\langle u, v \rangle}) = -2i \cdot 2i \operatorname{Im} \langle u, v \rangle = 4 \operatorname{Im} \langle u, v \rangle\end{aligned}$$

vlastnost kompl. č. $\left\{ \begin{array}{l} x - \bar{x} = a+bi - (a-bi) = 2bi = 2i \operatorname{Im}(x) \end{array} \right.$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

